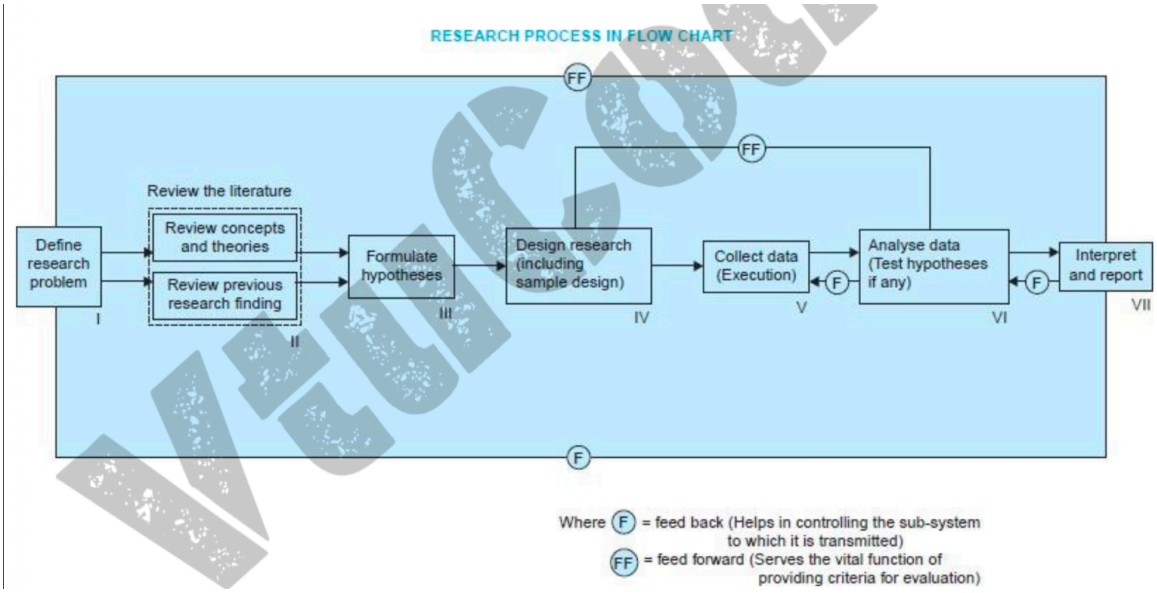
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1. Define research and explain the research flow cycle with a relevant diagram.

Definition of Research

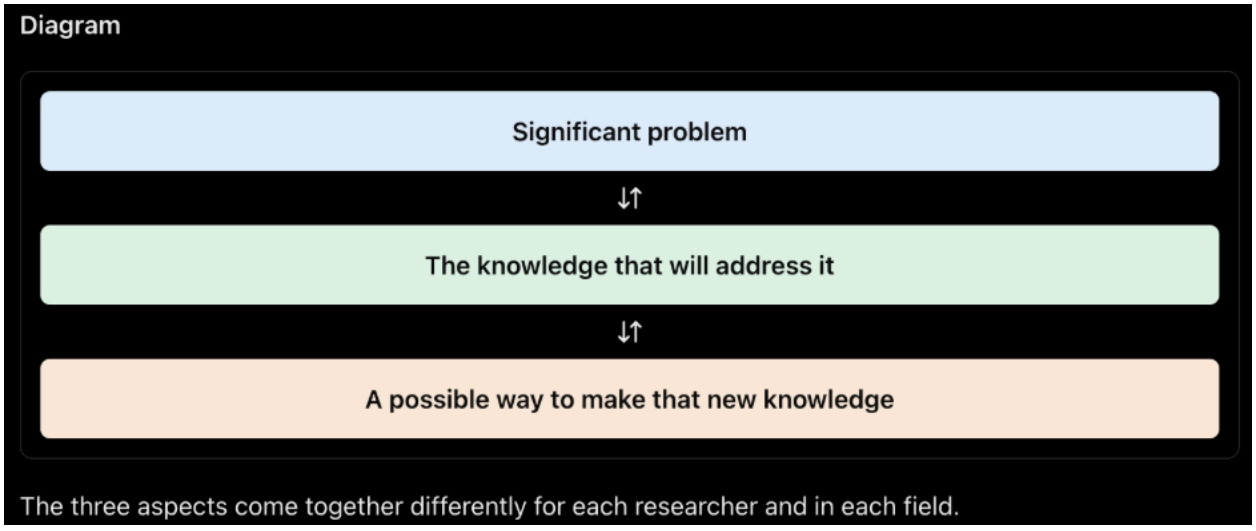
- Research refers to a careful, well-defined (or redefined), objective, and systematic method of search for knowledge, or formulation of a theory that is driven by inquisitiveness for that which is unknown and useful on a particular aspect so as to make an original contribution to expand the existing knowledge base.
- Research involves formulation of hypothesis or proposition of solutions, data analysis, and deductions; and ascertaining whether the conclusions fit the hypothesis.
- Research is a process of creating, or formulating knowledge that does not yet exist. Thus research is an art of scientific investigation.
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Research Flow Cycle

The research process consists of a series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps.

1. Formulating the research problem: The researcher must single out the problem to be studied and decide the general area of interest or aspect of a subject to inquire into.
2. Extensive literature survey: The researcher undertakes an extensive literature survey connected with the research problem.
3. Development of working hypotheses: The researcher states the working hypothesis or hypotheses in clear terms.
4. Preparing the research design: The researcher prepares the conceptual structure within which the research would be conducted.
5. Determining sample design: The researcher decides the way of selecting a sample from a given population.
6. Collecting the data: The researcher collects appropriate data through suitable methods.
7. Execution of the project: The project is executed systematically and on time.
8. Analysis of data: The collected data are classified, coded, tabulated, and analysed.
9. Hypothesis-testing: The researcher tests whether the facts support or contradict the hypothesis.
10. Generalizations and interpretation: If a hypothesis is tested and upheld several times, the researcher may arrive at generalizations and build a theory.
11. Preparation of the report or thesis: Finally, the researcher prepares the report of the work completed

2.What are the three broad categories of developing and accessing knowledge in research? Explain



The three parts that must be brought together in research are:

1. Significant problem
- A good research objective must address a significant problem.
- The problem should be original, significant, and solvable or achievable.
2. The knowledge that will address it
- The significance, originality, theory, tools, and methods needed to address a problem normally come from existing recorded literature and knowledge in the field.
- The researcher must be aware of what already exists in the field.
3. A possible way to make that new knowledge
- The researcher must identify a possible method and the tools that could be used to obtain new knowledge.
- A good research objective should be achievable.

3.What are three ways to credit the research contributions? Explain

The three ways to credit research contributions are:

1. Authorship: Authorship of the intended publication.
2. Citation: Citation of previously published or formally presented work.
3. Written acknowledgment: Acknowledgment of some inputs to the present research.

Explanation

- Authorship: Establishes accountability and gives due credit. A person is expected to be listed as an author when they significantly contribute to research design, data interpretation, or writing of the paper.
- Citation: Gives credit to previously published work and allows readers to trace the original source.
- Written acknowledgment: Recognizes contributions from people or organizations that helped with the research but may not qualify for authorship.

4. What are the objectives of Engineering Research?

- To solve new and important problems: The conclusion at the end of a research outcome has to be new, but when one starts, the conclusion is unknown.

- To solve engineering problems: Knowing where and how to find different types of information helps solve engineering problems in academic and professional careers.
- To support decision-making: The ability to conduct thorough and accurate research while clearly communicating the results is extremely important in decision-making.
- To seek answers to open questions: The main aim of research is to apply scientific approaches to seek answers to open questions.
- To develop new knowledge: The objectives of engineering research should be to develop new theoretical or applied knowledge and not necessarily be limited to obtaining abilities to obtain the desired result.
- To understand unsuccessful results: If the desired result cannot be achieved, the researcher should understand why it is not possible, because that is also a contribution toward ongoing research in solving that problem.

5.Explain the importance of motivation in engineering research.

1. Intrinsic motivation: Interest, challenge, learning, meaning, and purpose are linked to strong creative performance.
2. Extrinsic motivation: Rewards for good work include money, fame, awards, praise, and status. These are strong motivators but may block creativity.
3. Influence of others: Competition, collaboration, commitment, and encouragement from others are motivating factors in research.
4. Personal motivation: Solving unsolved problems, intellectual joy, service to the community, and respectability are driving factors.
5. Improving technology: Wanting to do better than what has been achieved in the world and improving the state of the art in technology motivate researchers.
6. Improving society: Contributing to the improvement of society and fulfilling the historical legacy in the immediate socio-cultural context are motivating factors.
7. Other factors: Government directives, funding opportunities in certain areas, and terms of employment can motivate people to get involved in engineering research.

6.Discuss the different types of engineering research. Clearly point out the differences between all of them

with examples.

1. Descriptive versus Analytical Research

Descriptive Research

- Includes comparative and correlational methods and fact-finding inquiries to describe the present state of the art.
- The researcher does not control the variables but reports them as they are.
- It also includes attempts to determine causes even though the variables cannot be controlled.

Analytical Research

- Uses already available facts for analysis and critical evaluation.
- Some research studies can be both descriptive and analytical.

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2. Applied versus Fundamental Research

Applied Research

- Seeks to solve an immediate problem facing an organization.
- Its primary objective is to find a solution to compelling problems in actual practice.

Fundamental Research

- Also called basic or pure research.
- Is concerned with generalizations and formulation of a theory.
- Aims to seek information that could have a broad base of applications in the medium to long term.
- Research concerning natural phenomena or pure mathematics are examples.

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3. Quantitative versus Qualitative Research

Quantitative Research

- Uses statistical observations of a sufficiently large number of representative cases to draw conclusions.

Qualitative Research

- Relies on a few non-representative cases or verbal narratives in behavioural studies to make a proposition.

- Example given in the notes: studying the clustering effect at intersections in transportation engineering.

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4. Conceptual versus Empirical Research

Conceptual Research

- Relates to abstract ideas or theories.
- Is generally used by philosophers and thinkers to develop new concepts or reinterpret existing ones.

Empirical Research

- Relies on experience or observation.
- It is data-based research, producing conclusions that can be verified by observation or experiment.
- The researcher forms a working hypothesis and collects sufficient facts or data to prove or disprove it.

7. List the different types of Research Misconduct. Provide examples for each.

1. Fabrication (Illegitimate creation of data)

- Fabrication is the act of creating false data or experiments instead of obtaining actual results.
- It may happen due to timeline pressures from supervisors or customers.
- Example: Creating experimental results without actually conducting the experiment.

2. Falsification (Inappropriate alteration of data)

- Falsification is the misrepresentation, misinterpretation, or illegitimate alteration of data or experiments.
- It is done to support a desired hypothesis even when the actual experimental data suggest otherwise.
- Example: Altering experimental data to make it support the desired hypothesis.

3. Plagiarism (Taking others' work without attribution)

- Plagiarism occurs when someone uses or reuses another person's work, such as text, data, tables, figures, illustrations, or concepts, as their own without explicit acknowledgment.
- Verbatim copying or reusing one's own published work is termed self-plagiarism.
- Example: Copying another researcher's text into a research paper without acknowledging the original source.

4. Other aspects of research misconduct

- Serious deviations from accepted conduct can be considered research misconduct.
- When deception and damage occur together, a fraud is deemed to have taken place.
- Example: Submitting the same article simultaneously to two different journals, which violates publication policies.

8.Give a detailed description of Ethics and Ethical practices in Research.

A. Ethics in Engineering Research

1. Meaning of Ethics: Ethics refers to a set of rules distinguishing acceptable and unacceptable conduct, and distinguishing right from wrong.

2. Ethical Norms: Most people learn ethical norms during their formative years, but moral development continues through different stages of growth. There may be differences in the interpretation and application of common ethical norms.

3. Ethics and Laws: Ethical principles can be used for evaluation, proposition, or interpretation of laws. Ethics are not laws, but laws often follow ethics because ethics are our shared values.

4. International Research Ethics: International norms for the ethical conduct of research have existed since the adoption of the Nuremberg Code in 1947.

5. Research Credit: According to Whitbeck, issues related to research credit date back to the establishment of the British Royal Society (BRS) in the seventeenth century. The BRS gave priority to whoever first submitted findings for publication.

B. Ethical Practices in Engineering Research

1. Research Ethics and Responsible Conduct: Research ethics examines the appropriate application of research outcomes, while responsible conduct of research deals with the way the work is undertaken.

2. Ethical Decisions: Engineering researchers need to make ethical decisions regarding technological developments, privacy issues, and data related to surveillance systems. They are answerable for the repercussions of their research outcomes.

3. Ethics in Data: Ethics matter in data used in engineering research because there is usually an impact on humans. Researchers must consider whether certain uses of data are acceptable.

4. Engineering Ethics: Engineering ethics provides rules for deciding what is acceptable and what is not. Engineering research is not separate from technological development.

5. Setting Ethical Requirements: By setting ethically correct requirements at the beginning, engineering researchers can influence the effects of the developed technology.

6. Ethical Design: During the design process, decisions must be made about the priority of requirements while considering ethical aspects.

7. Choosing Alternatives: Engineering researchers have to choose between different alternatives that fulfil similar functions.

8. Minimizing Risks: Researchers have an ethical responsibility to minimize hazards and risks associated with the technologies they develop and consider safer alternatives.

9. Safety in Design: Designs should be made inherently safe wherever possible. They should include safety factors, multiple independent safety barriers, or a supervisory mechanism to take control if the primary process fails.

9.What are the key ethical issues related to authorship? Explain each one.

1. Who should be included as an author?
- A person should be listed as an author only when they make a significant contribution to research design, data interpretation, or writing of the paper.
2. Order of authors
- The appropriate order of listing authors is an important issue in research. The order should recognize the contributions of the researchers.
3. Co-authorship and contributions

- Some people may be deeply involved in conducting research but may not contribute to drafting the paper. Their contributions and appropriate recognition must be considered.

4. Consent of authors

- All authors must agree to the content of the research paper and give their consent for its submission.

5. Accountability of co-authors

- All listed authors have responsibility for the contents of the research article. The contributions of each co-author should be identified to determine their appropriate accountability.

6. Double submission

- Double submission involves submitting the same paper to two publication forums simultaneously. Reputed journals want to publish original papers and strongly discourage this practice.

10.Differentiate between basic research and applied research in engineering.

Basic Research	Applied Research
1. Also called **fundamental or pure research.**	1. Seeks to solve an immediate problem facing an organization.
2. It is concerned with generalizations and formulation of a theory.	2. Its primary objective is to find a solution to problems in actual practice.
3. It seeks information that could have a broad base of applications in the medium to long term.	3. It focuses on solving practical problems.
4. **Example:** Research concerning natural phenomena or pure mathematics.	4. **Example:** Research to identify social or economic trends, or to find out whether certain communications will be read and understood.

11. Explain the different types of research misconduct and their consequences.

1. Fabrication (Illegitimate creation of data)

- Fabrication is the act of creating false data or experiments without actually obtaining the results.

- It may occur due to timeline pressures from supervisors or customers.

Consequences:

- Hampers engineering research.
- Causes false empirical data to enter the literature.
- Damages the trustworthiness of the individuals involved.
- Incurs additional costs and impedes research progress.
- Causes avoidable delays in technical advancement.

2. Falsification (Inappropriate alteration of data)

- Falsification is the misrepresentation, misinterpretation, or illegitimate alteration of data or experiments to support a desired hypothesis, even when the actual experimental data suggest otherwise.

Consequences:

- Hampers engineering research.
- Introduces misleading results into the literature.
- Damages trustworthiness.
- Incurs additional costs and delays research progress and technical advancement.

3. Plagiarism (Taking others' work without attribution)

- Plagiarism occurs when someone uses another person's work, such as text, data, tables, figures, illustrations, or concepts, as their own without proper acknowledgment.
- Copying or reusing one's own published work is called self-plagiarism.

Consequences:

- It is an unacceptable practice in scientific literature.
- It misrepresents another person's work as one's own.
- It can be detected by original authors, reviewers, editors, or readers.

4. Other Aspects of Research Misconduct

- Serious deviations from accepted research conduct can be considered research misconduct.
- When deception and damage occur together, fraud is deemed to have taken place.
- Simultaneous submission of the same article to two different journals violates publication policies.
- Mistakes in published articles may not be reported or corrected.

Consequences:

- Ethical violations may eventually be exposed.
- Research integrity and publication practices are affected.